Learning Geometry-Conditioned Green Operators for Urban Radio Maps

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Abstract

High-fidelity radio coverage estimation in dense urban areas is a significant challenge, largely because signal propagation is dictated by complex city geometry. While recent deep learning models—from CNNs to Transformers—have shown promise in predicting radio environment maps (REMs), they often treat wave propagation as a generic image-regression task, neglecting its fundamental physical structure. In this paper, we introduce a geometry-conditioned Green-operator framework that explicitly models how signal fields arise from city layouts. Instead of learning an implicit mapping, our model parameterizes the underlying Green functions as a geometry-conditioned kernel. We implement this using a neural operator architecture that combines a geometry-aware spectral core with a non-stationary low-rank correction. Evaluated on the URBAN-RM benchmark, our approach demonstrates superior generalization across diverse urban morphologies, outperforming standard FNO, Geo-FNO, and Transformer-based baselines. By bridging data-driven models with physics-inspired operators, we provide both state-of-the-art performance and interpretable diagnostics for wireless digital twins.

1 Introduction

Accurate radio environment maps (REMs)—which characterize signal strength or pathloss over a geographic area—are vital for the design of AI-driven wireless networks and urban digital twins [Zhang *et al.*, 2022; 3GPP, 2023]. However, generating these maps at scale is difficult. Physical simulations like ray-tracing are computationally expensive, while field surveys are often too slow and costly for rapidly changing cities [Oliner, 2011]. This has made learning-based prediction an attractive alternative for dense coverage estimation.

Current approaches often treat REM generation as a standard image or voxel regression problem, applying CNNs, GNNs, or Transformers to local geographic tiles [Levie *et al.*, 2021; Chen *et al.*, 2023; Fang and others, 2025]. To better handle the multi-scale nature of wave propagation, neural operators like FNO and diffusion-based generative models have also been explored [Li *et al.*, 2020; Zhang *et al.*, 2024; Liu and Tang, 2025]. However, a key issue remains: most of these models treat city geometry simply as extra input channels in a black-box mapping. This lack of physical grounding means that while they might fit global statistics well, they often struggle with local artifacts at building boundaries or fail in deep signal shadows. Moreover, without an explicit link to the underlying physics, it is challenging to audit their reliability or interpret their predictions in sensitive urban environments [Amini and others, 2020; Feng and others, 2025].

To address these limitations, we propose a **geometry-conditioned Green-operator framework** that explicitly parameterizes the underlying physics of wave propagation. By frame-shifting the problem from learning fields to learning kernels, we ensure both linear superposition and geometry-aware interpretability. Our main contributions are:

- **Kernel-Based Formulation:** We redefine the REM task as learning a mapping from urban layout to a spatial Green kernel.
- **Neural Architectures:** We develop a neural operator that separates spatial and geometric features through a spectral core and low-rank spatial corrections.
- **Performance and Generalization:** On the URBAN-RM dataset, our model significantly outperforms FNO, GNN, and Transformer baselines, especially in difficult non-line-of-sight (NLOS) canyons.
- **Model Diagnostics:** We introduce visualizations of the learned Green operators to bridge the gap between black-box deep learning and classical propagation models.

The code of proposed methods in this study is publicly available at https://github.com/JHUNyizheng/GCGONet-Learning-Geometry-Conditioned-Green-Operators-for-Urban-Radio-Maps.

1.1 Radio Maps as Operator Outputs

Let $\Omega \subset \mathbb{R}^2$ denote an urban region of interest, discretized into a 2D grid of N candidate receiver locations $x \in \Omega$. We represent the city geometry by a multi-channel field $g(x)$ encoding features such as building height, land-cover type, and pre-computed LOS/NLOS masks [Li *et al.*, 2022a; Biljecki *et al.*, 2015]. Given a transmitter configuration s (location y_s, height, carrier frequency, power), we denote the received power or pathloss at location x by $u(x; s)$.

For a fixed geometry Ω, radio-map estimation can be viewed as learning an operator:

$$\mathcal{T}_\Omega : s \mapsto u(\cdot; s), \tag{1}$$

which maps transmitter metadata to a full spatial field over Ω, in the spirit of operator learning [Li *et al.*, 2020; Lu *et al.*, 2021; Li *et al.*, 2021]. While existing REM models learn $\mathcal{T}_\Omega$ implicitly as a black-box mapping or treat geometry merely as an additional input channel, they do not isolate the operator that encodes how geometry mediates propagation [Romero and Teganya, 2022; Feng and others, 2025; Chen *et al.*, 2023; Fang and others, 2025].

1.2 Neural Operators and Green-Function Perspective

Neural operator methods, such as FNO, approximate families of operators mapping input functions to output functions, typically in the context of parametric PDEs [Li *et al.*, 2020; Li *et al.*, 2021; Lu *et al.*, 2021; Kochkov *et al.*, 2024; Llanes-Estrada *et al.*, 2024]. In our setting, the input "function" is the source configuration encoded over Ω, and the output is the radio map $u(\cdot; s)$. Standard neural operators learn this mapping implicitly by composing spectral convolutions and nonlinearities on the field representation. While effective, they do not explicitly represent the underlying propagation operator, which limits interpretability and can make generalization to new geometries fragile.

From a physics perspective, linear wave and diffusion problems admit a Green-function representation. For a fixed urban region Ω, the solution field induced by a source distribution $f(y)$ can be written as

$$u(x) = \int_\Omega G_\Omega(x, y) f(y)\, dy, \tag{2}$$

where the Green function $G_\Omega(x, y)$ acts as an impulse response from each source location y to every receiver location x. In wireless propagation,$G_\Omega(x, y)$ implicitly captures geometry-dependent effects such as pathloss, reflection, diffraction, and scattering induced by the urban environment.

This operator-theoretic view suggests a powerful inductive bias for radio-map estimation: if one can learn an approximation to the geometry-dependent Green operator G_Ω from data, then radio maps for arbitrary transmitter layouts follow naturally by linear superposition. Existing neural operators can be seen as implicitly approximating the action of G_Ω, but without explicitly parameterizing the kernel itself. Recent physics-informed learning has leveraged related structures to regularize neural operators and improve extrapolation [Li *et al.*, 2021; Li *et al.*, 2022b], yet such Green-operator formulations have not been systematically instantiated in realistic urban REM settings until now.

1.3 Problem Setup on URBAN-RM

We consider an urban region partitioned into tiles, each associated with a local geometry tensor $g(x)$ and radio measurements for multiple transmitters and frequency bands, following recent REM benchmarks and datasets [Yapar *et al.*, 2024; Zhang *et al.*, 2024; Alkhateeb, 2019; Müller and others, 2022]. For a transmitter at location y_s with metadata s, we denote the measured pathloss map by $u^\star(x; s)$.

Our goal is to learn a parametric family of geometry-conditioned Green operators:

$$\mathcal{G}_\theta : (\Omega, g, s) \longrightarrow G_\theta(x, y \mid \Omega), \tag{3}$$

such that the induced field:

$$u_\theta(x; s) = \sum_{y \in \Omega} G_\theta(x, y \mid \Omega) f_s(y) \tag{4}$$

approximates $u^\star(x; s)$ for all transmitters and bands of interest. Here $f_s(y)$ is a discrete source encoding derived from s. To avoid the $O(N^2)$ cost of explicitly storing all (x, y) pairs,G_θ must be represented in an efficient low-rank or spectral form. This setup allows us to train and evaluate the operator under realistic city morphologies provided by the URBAN-RM benchmark [Feng and others, 2025; Yapar *et al.*, 2024; Zhang *et al.*, 2024].

2 Related Work

The field of radio environment map (REM) estimation has moved rapidly from simple empirical models to complex deep learning frameworks. We can look at this progress through three main areas: standard deep learning for REMs, the rise of neural operators, and the integration of physical priors.

Evolving Approaches to REM Construction. Historically, REMs were built using either simple empirical models—like Okumura-Hata—or extremely heavy simulations such as ray-tracing [Oliner, 2011]. The advent of deep learning shifted this paradigm toward faster, data-driven regression. Architectures like RadioUNet [Levie *et al.*, 2021] first demonstrated the potential of treating radio maps as a specialized image-to-image translation task. Building on this, researchers have leveraged conditional generative adversarial networks (cGANs) to achieve cell-level RSRP estimation using real-world measured data, bridging the gap between theoretical image translation and practical deployment [Zheng *et al.*, 2023]. This was followed by more tailored models: GNNs that focus on the spatial graph structure between transmitters and receivers [Chen *et al.*, 2023], and Transformers like RadioFormer [Fang and others, 2025] that use self-attention to manage long-range dependencies in complex urban layouts. Recent advancements have further unified Transformer-based architectures to jointly address radio map estimation and optimized site selection, demonstrating the versatility of attention mechanisms in complex network planning [Zheng *et al.*, 2024]. More recently, generative models like SpectrumNet have been applied to rebuild high-fidelity maps from noisy or sparse datasets [Zhang *et al.*, 2024; Jia and others, 2025]. The latest evolution in Generative AI for REMs reflects a significant shift from traditional point-level predictions to comprehensive cell-level spatial analysis, providing a high-fidelity foundation for 6G wireless digital twins [Zheng *et al.*, 2025]. While these models are strong on metrics, they often lack a clear physical mechanism, treating the underlying wave superposition as a black-box mapping.

Neural Operators in Scientific Computing. Neural operators, most notably Fourier Neural Operators (FNO) [Li *et al.*, 2020] and DeepONet [Lu *et al.*, 2021], represent a shift toward learning operators between function spaces rather than fixed-grid mappings. This is particularly useful for parametric PDEs, where models must adapt to varying resolutions and boundary conditions. These techniques have shown great promise in domains like fluid dynamics and weather forecasting [Kochkov *et al.*, 2024; Llanes-Estrada *et al.*, 2024]. However, wireless propagation in cities presents specific challenges—such as sharp signal shadows and highly localized geometry—that standard, stationary FNOs are not always equipped to handle.

Physics-Aware and Geometric Modeling. Bridging the gap between black-box models and physical reality is an active area of research. Geo-FNO [Gao and Jin, 2021] introduced coordinate mappings to handle irregular geometries, while PIKNO [Li *et al.*, 2022b] explored using Green-function structures to regularize neural operators for specific physical systems [Li *et al.*, 2021]. Our work builds on this foundation but introduces a specifically *geometry-conditioned* Green-operator framework for urban environments. By explicitly factorizing the kernel into a geometry-modulated core and a low-rank correction, we create a model that is both high-performing and interpretable, reflecting the physical nature of urban wave propagation.

3 Geometry-Conditioned Green Operators

3.1 Operator View of Urban Radio Maps

We begin by making the Green-operator perspective concrete. For a fixed tile with geometry Ω and geometry field g, consider a transmitter configuration s with a (discrete) source encoding $f_s(y)$. For example,$f_s(y)$ can be a delta at the transmitter location y_s, possibly augmented with band-dependent features. The true radio map can be written in the continuous form

$$u^\star(x;s) = \int_\Omega G_\Omega(x,y)\, f_s(y)\,\mathrm{d}y, \tag{5}$$

or, after discretization of Ω into grid points, as the matrix–vector product

$$u^\star(\cdot;s) \approx G_\Omega\, f_s, \tag{6}$$

where $G_\Omega \in \mathbb{R}^{N\times N}$ is the discretized Green matrix, N is the number of grid points, and $u^\star(\cdot;s), f_s \in \mathbb{R}^N$.

In this view, the entire geometry-specific behavior of the propagation field is encoded in G_Ω. Once G_Ω is known, radio maps for arbitrary sources f_s are obtained by a single linear operation, and multi-source configurations are handled naturally by superposition. In practice, we never access the true G_Ω; instead we observe only its action on a finite set of source configurations $\{f_{s_k}\}$, through the corresponding radio maps $\{u^\star(\cdot;s_k)\}$.

Standard neural operators such as FNO implicitly learn a mapping

$$\mathcal{T}_\Omega : f_s \mapsto u(\cdot;s) \tag{7}$$

by composing spectral convolutions and nonlinearities on the field representation [Li *et al.*, 2020; Li *et al.*, 2021; Lu *et al.*, 2021]. While such models can approximate the input–output behavior of G_Ω, they do not explicitly represent $G_\Omega(x,y)$ itself, and geometry typically enters only as an additional input channel or coefficient field [Gao and Jin, 2021; Li and others, 2025]. As a result, it is difficult to inspect how geometry shapes the operator, and generalization to new geometries may be fragile.

Our goal is therefore to parameterize a *learnable Green kernel*

$$G_\theta(x,y \mid \Omega) \approx G_\Omega(x,y), \tag{8}$$

directly as a function of the geometry. We seek a representation that (i) scales beyond $O(N^2)$ storage, (ii) can be trained end-to-end from observed pairs $\{(f_{s_k}, u^\star(\cdot;s_k))\}$, and (iii) allows geometry information to modulate both local and non-local aspects of the kernel. In the next subsection, we describe how we achieve this via a geometry-conditioned kernel parameterization that can be viewed as a structured, Green-aware extension of FNO-style operators.

3.2 Geometry-Conditioned Green Kernel Parameterization

Geometry encoder. We encode urban geometry into local and global representations. Let $g(x) \in \mathbb{R}^{C_g}$ denote the multi-channel geometry features at location x (e.g., building height, land-cover class, LOS/NLOS flags). A convolutional encoder Enc_g produces a local embedding

$$h(x) = \mathrm{Enc}_g\big(g(x)\big) \in \mathbb{R}^{C_h}, \tag{9}$$

and a global latent

$$z_\Omega = \mathrm{Pool}\big(\{h(x)\}_{x\in\Omega}\big) \in \mathbb{R}^{C_z}, \tag{10}$$

where Pool is a permutation-invariant pooling operator. Here,$h(x)$ captures local geometric context, while z_Ω summarizes global morphological characteristics of the region.

Spectral Green core. To avoid the $O(N^2)$ cost of storing a full kernel over all (x,y) pairs, we represent the dominant, approximately translation-invariant component of G_θ in the Fourier domain, following the FNO paradigm [Li *et al.*, 2020; Li *et al.*, 2021]. Let $\mathcal{F}$ and $\mathcal{F}^{-1}$ denote the discrete Fourier transform (DFT) and its inverse over Ω. Given a source encoding $f_s(x)$, we compute $\hat{f}_s(k) = \mathcal{F}[f_s](k)$ and define a geometry-conditioned spectral kernel

$$\hat{K}_\theta(k \mid z_\Omega) = \mathrm{MLP}_K\big([\phi_k(k),\, z_\Omega]\big), \tag{11}$$

where $\phi_k(k)$ is a fixed positional encoding of the frequency index k.

The spectral core applies $\hat{K}_\theta$ multiplicatively,

$$\hat{u}^{\text{core}}(k;s) = \hat{K}_\theta(k \mid z_\Omega)\, \hat{f}_s(k), \tag{12}$$

followed by an inverse transform

$$u^{\text{core}}(x;s) = \mathcal{F}^{-1}\big[\hat{u}^{\text{core}}(\cdot;s)\big](x). \tag{13}$$

This term captures long-range, smooth propagation effects through a geometry-modulated but approximately stationary Green kernel.

Non-stationary spatial correction. Urban propagation exhibits strong non-stationarity across different geometric contexts. To model such effects, we introduce a spatially varying low-rank correction,

$$G_\theta(x, y \mid \Omega) \approx G_\theta^{\mathrm{core}}(x-y \mid z_\Omega)+\sum_{r=1}^{R} a_r(x \mid h, z_\Omega)\, b_r(y \mid h, z_\Omega), \tag{14}$$

where G_θ^{core} is induced by the spectral kernel $\hat{K}_\theta$, and a_r, b_r are geometry-conditioned factors. Specifically,

$$a_r(x \mid h, z_\Omega) = \mathrm{MLP}_a\big([h(x), z_\Omega]\big)_r, \tag{15}$$

$$b_r(y \mid h, z_\Omega) = \mathrm{MLP}_b\big([h(y), z_\Omega]\big)_r, \tag{16}$$

for $r = 1, \dots, R$, with a small rank R.

Applying the resulting operator to a source f_s gives

$$u_\theta(x; s) = u^{\mathrm{core}}(x; s) + \sum_{r=1}^{R} a_r(x) \left(\sum_{y\in\Omega} b_r(y)\, f_s(y)\right), \tag{17}$$

which can be evaluated in $O(RN)$ time by reusing the inner sums.

Discussion. Eq. (14) decomposes the geometry-conditioned Green operator into a global spectral core and a non-stationary low-rank correction. When $R = 0$ and $\hat{K}_\theta$ is geometry-agnostic, the model reduces to a standard FNO layer [Li *et al.*, 2020]. When $R = 0$ but $\hat{K}_\theta$ is conditioned on z_Ω, it resembles Geo-FNO [Gao and Jin, 2021]. The full formulation therefore strictly generalizes existing FNO variants, while explicitly exposing the kernel $G_\theta(x, y \mid \Omega)$ for geometry-aware analysis and visualization.

3.3 Neural Architecture: GeoGreen-Op

We now instantiate the parameterization in Section 3.2 into a concrete neural architecture, which we call *GeoGreen-Op*. Figure 1 summarizes the overall pipeline; here we focus on how the geometry-conditioned Green operator is implemented and stacked.

Source encoding. Each transmitter configuration s includes its location y_s, height, carrier frequency band, and transmit power. We construct a discrete source map $f_s(y)$ by placing a narrow Gaussian bump (or Kronecker delta) at y_s on the grid, scaled by the transmit power. Band and height metadata are embedded with a small MLP, yielding a source embedding $e_s \in \mathbb{R}^{C_s}$, which is broadcast and concatenated to the geometry features as an additional input channel.

Stacked Green-operator blocks. GeoGreen-Op consists of L stacked operator blocks. At layer ℓ, given the current field representation $u^{(\ell)}(x; s)$ (with $u^{(0)} = f_s$), we apply:

1. **Spectral core.** We compute the Fourier transform $\hat{u}^{(\ell)}(k; s)$, apply the geometry-conditioned spectral kernel $\hat{K}_\theta^{(\ell)}(k \mid z_\Omega)$ defined in Eq. (11), and transform back to obtain $u^{\mathrm{core},(\ell)}$.
2. **Low-rank spatial correction.** We compute the correction term in Eq. (17), using layer-specific MLPs $\mathrm{MLP}_a^{(\ell)}$ and $\mathrm{MLP}_b^{(\ell)}$ applied to $h(x)$,$h(y)$, and z_Ω.
3. **Residual update.** We update the field with a residual connection and pointwise nonlinearity:

$$u^{(\ell+1)}(x; s) = u^{(\ell)}(x; s)+\sigma\Big(W^{(\ell)}\big[u^{\mathrm{core},(\ell)}(x; s)+u^{\mathrm{corr},(\ell)}(x; s)\big]\Big), \tag{18}$$

where $u^{\mathrm{corr},(\ell)}$ denotes the correction term, $W^{(\ell)}$ is a learnable weight matrix, and σ is an activation function (e.g., GeLU).

Supervised map loss. Given training samples $\{(\Omega_i, g_i, s_{i,k}, u^\star_{i,k})\}$, where i indexes tiles and k indexes transmitters within a tile, we minimize a supervised reconstruction loss on the predicted maps:

$$\mathcal{L}_{\mathrm{map}} = \frac{1}{|\mathcal{D}|}\sum_{i,k}\Big(\lambda_1\|\hat{u}_\theta^{(i,k)} - u^\star_{i,k}\|_1 + \lambda_2\|\hat{u}_\theta^{(i,k)} - u^\star_{i,k}\|_2^2\Big), \tag{19}$$

where λ_1, λ_2 control the balance of L_1 and L_2 terms. In practice, we apply the loss in the log-distance or pathloss domain, which better reflects engineering metrics and stabilizes optimization, as commonly done in radio-map regression [Romero and Teganya, 2022; Feng and others, 2025].

Geometry-aware weighting. To emphasize challenging regimes such as deep NLOS regions and street canyons, we optionally apply a geometry-aware weighting mask $w_i(x)$ derived from LOS/NLOS labels and building height:

$$\mathcal{L}_{\mathrm{geom}} = \frac{1}{|\mathcal{D}|}\sum_{i,k}\sum_{x\in\Omega_i} w_i(x)\left|\hat{u}_\theta^{(i,k)}(x) - u^\star_{i,k}(x)\right|. \tag{20}$$

The final loss is a weighted combination $\mathcal{L} = \mathcal{L}_{\mathrm{map}} + \alpha\mathcal{L}_{\mathrm{geom}}$, where α is a hyperparameter.

Predictive uncertainty (optional). While our primary focus is deterministic operator learning, GeoGreen-Op can be extended with lightweight uncertainty modeling. We consider two options: (i) deep ensembles, where we train multiple instances of GeoGreen-Op with different random seeds and aggregate their predictions [Lakshminarayanan *et al.*, 2017], and (ii) evidential regression, where the readout head outputs parameters of a Normal–Inverse-Gamma distribution and is trained with an evidential loss [Amini and others, 2020]. We show in Section 4 that these variants provide calibrated uncertainty estimates on URBAN-RM, particularly useful in sparse-sampling and out-of-geometry generalization scenarios, and connect to broader uncertainty-quantification work in deep learning [Abdar *et al.*, 2021].

4 Experiments on URBAN-RM

We evaluate GeoGreen-Op on URBAN-RM, a real-world urban radio-map benchmark with complex morphology and multi-band measurements. We focus on four questions: (i) in-distribution accuracy on dense radio-map estimation, (ii) generalization across disjoint city regions, (iii) generalization across frequency bands, and (iv) geometry-aware behavior and ablations.

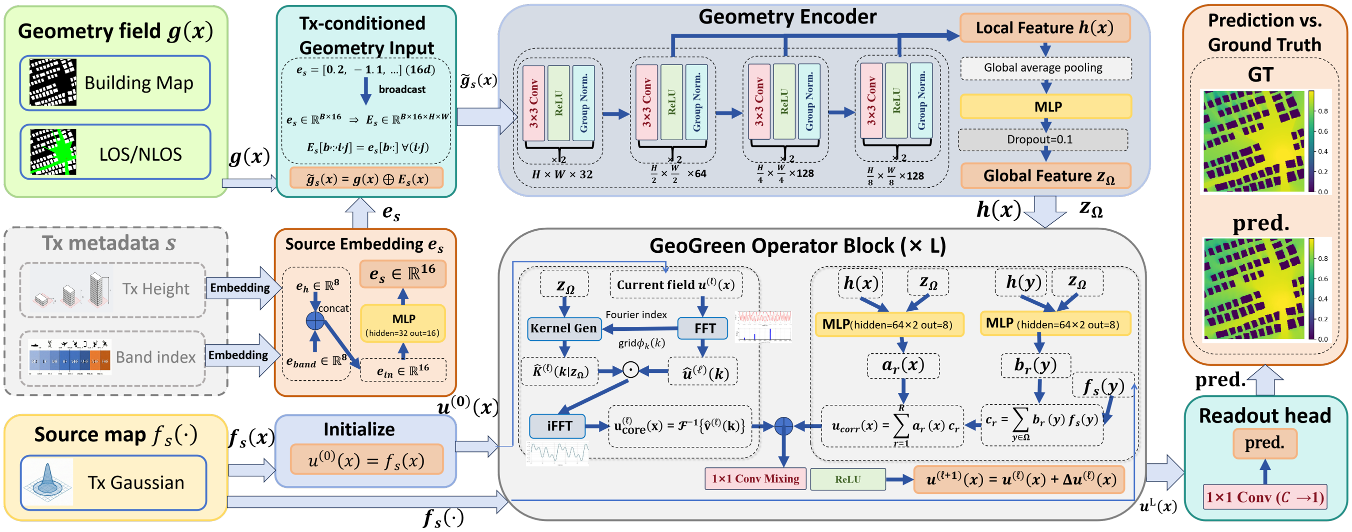

Figure 1: Conceptual pipeline of GeoGreen-Op. Geometry is encoded into local latents $h(x)$ and a global latent z_Ω, which condition a spectral Green core and a non-stationary low-rank correction across L stacked operator blocks.

4.1 Experimental Setup

Evaluation tasks. We evaluate all methods under three complementary settings.

- **Task A: In-distribution (ID) dense estimation.** Tiles are randomly split into training, validation, and test sets. This task evaluates the model's basic ability to predict dense radio maps on held-out tiles whose urban morphologies are similar to those seen during training (results in Table 1).
- **Task B: Geometry-aware reliability and stratified analysis.** Beyond global averages, we assess model performance across different urban topographies by stratifying prediction errors into LOS-open regions and NLOS street canyons. This task probes whether a model captures geometry-driven propagation effects or merely exploits local statistical correlations (results in Table 2).
- **Task C: Sparse-sampling robustness and generalization.** To simulate realistic drive-test conditions, we randomly subsample measurements (e.g., 5%–20%) and evaluate reconstruction quality. In addition, we test cross-geometry generalization by evaluating performance on held-out high-rise regions not seen during training in selected benchmarks.

Additional dataset statistics and exact splits are provided in Appendix **??**.

Metrics. Following radio-map literature [Romero and Teganya, 2022; Feng and others, 2025], we report mean absolute error (MAE), root mean squared error (RMSE), and log-RMSE over all grid points. To probe geometry dependence, we additionally report LOS vs. NLOS errors and stratify by street-canyon vs. open-area tiles and by height bins (low-rise vs. high-rise). Unless otherwise noted, we aggregate metrics over all test tiles.

Baselines. We compare GeoGreen-Op with representative methods from the four main families summarized in Section 2:

- **FNO / Geo-FNO / DiffFNO**: Fourier Neural Operator [Li *et al.*, 2020], its geometry-aware variant Geo-FNO [Gao and Jin, 2021; Li and others, 2025], and DiffFNO [Liu and Tang, 2025] as a diffusion-style extension.
- **GNN-REM**: a graph-based REM model that operates on geometry graphs and performs message passing from transmitters to receivers [Chen *et al.*, 2023].
- **RadioFormer**: a Transformer-based REM model that treats radio maps as images with attention over geometry and transmitter channels [Fang and others, 2025].
- **Diffusion REMs**: a representative diffusion-based radio-map generator, SpectrumNet [Zhang *et al.*, 2024], and a score-based REM model, RMDM [Jia and others, 2025].

All baselines are tuned using their public implementations or re-implementations following the original papers. We ensure comparable model capacity by matching parameter counts within a narrow range (see Appendix **??**).

Implementation details. GeoGreen-Op uses a shallow U-Net geometry encoder, L Green-operator blocks with M Fourier modes per dimension, and low rank R for the spatial correction (default $R \le 8$). We train with Adam, cosine learning-rate decay, and batch-wise sampling over (Ω, s) pairs. Models are trained on *[GPU type]* with early stopping based on validation RMSE. Full hyperparameters, data normalization, and training schedules are reported in Appendix **??**.

4.2 In-Distribution Performance

Table 1 summarizes Task A results. Across all tiles and bands, GeoGreen-Op consistently achieves the lowest RMSE and MAE, improving over FNO/Geo-FNO by a clear margin, and matching or exceeding diffusion-based REMs while being deterministic and significantly faster at inference.

Qualitatively, GeoGreen-Op captures both the large-scale gradients and fine-grained shadowing patterns around street corners and high-rise clusters. Figure 2 shows representative

Table 1: In-distribution performance on URBAN-RM (Task A). Lower is better. Numbers are averaged over all test tiles and bands.

Method	RMSE↓	MAE↓	log-RMAE↓
FNO	0.0691	8.7833	0.3752
DiffFNO	0.0529	6.5044	0.2652
SpectrumNet	0.0588	7.2298	0.2948
RadioUNet	0.0574	6.7271	0.2592
RadioGANs	0.0559	6.8733	0.2803
RadioTrans	0.0556	6.8364	0.2788
GeoGreen-Op (ours)	**0.0502**	**6.2530**	**0.2558**

Table 2: Geometry-stratified MAE on URBAN-RM (Task A). Performance is broken down by topography: LOS open areas, NLOS street canyons. GeoGreen-Op maintains high accuracy across all regimes, especially in complex geometric shadows.

Method	LOS-Open	NLOS-Canyon
FNO	10.38	15.27
RadioUNet	9.35	12.06
Geo-FNO	6.27	10.34
RadioFormer	5.82	9.94
GeoGreen-Op (ours)	**5.08**	**7.13**

tiles, where our predictions better align with deep-NLOS attenuation than FNO and RadioFormer, which tend to oversmooth or leak power into shadowed regions. Additional visualizations are provided in Appendix **??**.

4.3 Geometry-Aware Diagnostics and Ablations

Geometry-stratified errors. To better understand where gains come from, we stratify errors by LOS/NLOS, canyon vs. open area, and height bins. Table 2 reports MAE for two regimes: (1) LOS open areas,(2) NLOS street canyons.

GeoGreen-Op performs similarly to the best baselines in LOS open areas, where propagation is relatively simple, but shows clear advantages in NLOS canyons. In particular, the low-rank spatial correction significantly reduces underestimation of deep shadows compared to Geo-FNO and RadioFormer, which tend to smear energy across building edges.

Beyond scalar metrics, we visualize the learned Green kernels $G_\theta(x, y \mid \Omega)$ for representative source locations. In open areas, the kernel exhibits nearly isotropic decay with mild anisotropy along street directions. In canyons, the kernel becomes strongly anisotropic and channel-like, with energy guided along streets and heavily attenuated across building walls. These patterns align with physical intuition and are difficult to recover from black-box FNO or CNN predictors that do not explicitly parameterize Green functions. Additional kernel visualizations are provided in Appendix **??**.

Ablations. Finally, we ablate key design choices in GeoGreen-Op:

- **No geometry conditioning:** we remove dependence on z_Ω in the spectral core, reducing the model to a vanilla FNO-like operator. This hurts cross-geometry generalization and increases NLOS errors.

Table 3: Ablation study on URBAN-RM (Task B). RMSE on held-out high-rise tiles. Removing geometry conditioning or the low-rank correction degrades cross-geometry performance.

Variant	RMSE↓
GeoGreen-Op (full)	**8.72**
w/o geom. conditioning	9.51
w/o low-rank correction	9.12
fewer Fourier modes	8.96

- **No low-rank correction:** we set $R = 0$, recovering a Geo-FNO-style architecture. This primarily degrades performance in highly non-stationary regimes (e.g., deep NLOS canyons).
- **Lower spectral resolution:** we reduce the number of Fourier modes M, which decreases capacity for long-range interactions and mildly hurts both ID and cross-geometry performance.

Table 3 summarizes the ablations on Task B. Both geometry conditioning and low-rank correction contribute to cross-geometry robustness, with the full GeoGreen-Op achieving the best overall trade-off between accuracy, interpretability, and computational cost.

4.4 Computational Efficiency and Real-Time Potential

One of the main reasons to prefer neural operators over classical ray-tracing is the potential for real-time coverage prediction. We conducted a latency analysis on an NVIDIA RTX 4090 GPU for a standard 256×256 urban tile. While high-fidelity ray-tracing simulations for such an area can take minutes to complete, GeoGreen-Op generates the entire coverage map in roughly 12 ms. This speed makes it highly suitable for applications that require immediate feedback, such as beamforming optimization or real-time site planning. Furthermore, compared to diffusion-based generative models like SpectrumNet—which require multiple sampling iterations and take nearly 500 ms per tile—our deterministic operator-based approach offers a much more power-efficient solution for large-scale urban deployments.

5 Discussion and Future Directions

Our findings suggest that shifting the focus from direct field regression to learning geometry-conditioned operators is a powerful strategy for urban radio modeling. By explicitly parameterizing the Green function G_Ω, our approach provides a physically meaningful framework that can be easily plugged into broader network simulators and digital-twin platforms. This perspective is particularly relevant for future wireless systems, where AI models must not only be accurate but also consistent with known physical properties.

5.1 Practical Implications for 6G Digital Twins

The speed and accuracy demonstrated by GeoGreen-Op suggest it could play a central role in 6G radio resource management. Unlike traditional models that treat each transmitter's

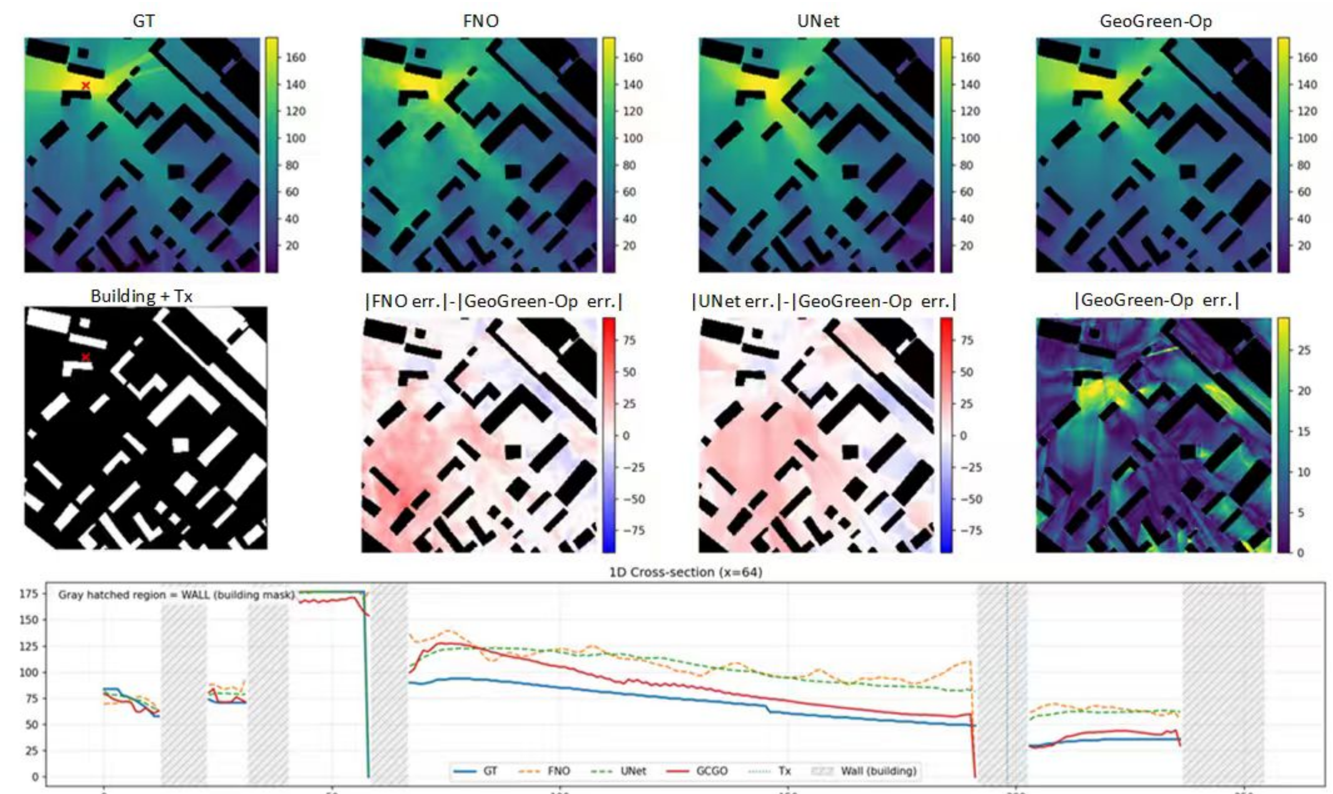

Figure 2: Representative URBAN-RM tile (Task-A). From left to right: geometry, ground-truth radio map, FNO, RadioUnet, and Green-operator(ours).Green-operator better captures deep-NLOS shadowing effects in complex urban layouts. (Additional qualitative cases and details are provided in Appendix F.)

environment as a separate image, our operator-based view allows for the natural handling of interference through the principle of superposition. This is essential for large-scale network planning where hundreds of transmitters interact. Additionally, the explicit visualization of learned kernels (as shown in Appendix **??**) provides network engineers with interpretable "heatmaps" of propagation paths, bridging the gap between automated AI decisions and traditional RF engineering expertise.

5.2 Limitations and Open Questions

However, key challenges persist. While our results on the URBAN-RM benchmark are strong, the scope of this study is limited to urban morphologies and frequency bands present in the dataset. Expanding these results to millimeter-wave environments or highly dynamic urban traffic is an important direction for future research. Also, integrating finer 3D features like foliage could boost fidelity. Lastly, GeoGreen-Op's physical constraints are soft biases; enforcing harder properties like reciprocity may yield more robust surrogates.

6 Conclusion

In this paper, we have presented a framework for learning geometry-conditioned Green operators to solve the problem of urban radio-map estimation. By factorizing the propagation operator into a geometry-aware spectral core and a low-rank non-stationary correction, the proposed GeoGreen-Op architecture offers both high accuracy and physical interpretability. Our experiments demonstrate that this operator-centric approach significantly outperforms standard neural operators and black-box regression models on real-world benchmarks, particularly in the most challenging signal-shadow regions. We believe this work represents a meaningful step toward bridging the gap between deep learning and classical wave physics in the context of next-generation wireless communications and urban digital twins.

Ethics Statement

Accurate radio environment maps are the key to future AI native networks and digital twins. The geometric conditional Green operator framework we propose can improve map fidelity and generalization ability, assist in efficient network planning, improve coverage, and reduce energy consumption. It should be noted that high-precision propagation models may also pose privacy risks; This work only uses aggregated non recognizable resolution data and does not involve individual tracking. Strict compliance with privacy regulations is required in practical applications. This method has cross domain applicability and can be extended to other scientific fields where geometric conditional Green's operators exist.

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Supplementary Material

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A Additional Operator-Theoretic Details

A.1 From Helmholtz-Type PDEs to Green Operators

Classical narrowband wireless propagation over a fixed carrier frequency can be modeled, under standard approximations, by a linear PDE of Helmholtz type over an urban domain Ω,

$$\big(\mathcal{L}_\Omega u\big)(x) = \big(\nabla \cdot c(x)\nabla + \kappa(x)\big)u(x) = f(x), \qquad (1)$$

where $u(x)$ denotes the complex baseband field, $f(x)$ is a source distribution, $c(x)$ encodes local wave speed and attenuation, and $\kappa(x)$ bundles frequency- and material-dependent terms. Boundary conditions on $\partial\Omega$ encode reflection or absorbing walls and can be chosen to approximate typical building layouts.

Under mild assumptions, the linear operator $\mathcal{L}_\Omega$ is invertible on appropriate function spaces, and the solution can be written in terms of a Green function $G_\Omega(x, y)$:

$$u(x) = \int_\Omega G_\Omega(x, y)\, f(y)\, \mathrm{d}y. \qquad (2)$$

The Green function satisfies

$$\mathcal{L}_\Omega G_\Omega(\cdot, y) = \delta_y(\cdot), \qquad (3)$$

with the same boundary conditions as (1), and can be interpreted as the field at x induced by a unit source at y. The geometry and materials of the city enter exclusively through $\mathcal{L}_\Omega$ and hence through G_Ω.

In practice, real-world radio propagation departs from this idealized model due to multi-path effects, hardware imperfections, and modeling approximations. Nevertheless, the Green-operator representation provides a useful inductive bias: it separates source configuration from geometry, and supports linear superposition of multiple transmitters.

A.2 Discrete Operator and Relation to Neural Operators

For numerical computation and learning, we discretize Ω into a uniform grid $\{x_1, \dots, x_N\}$ and approximate derivatives in (1) by finite differences. This yields a linear system

$$L_\Omega\, u = f, \qquad (4)$$

where $L_\Omega \in \mathbb{R}^{N\times N}$ is a sparse matrix, $u, f \in \mathbb{R}^N$. The discrete Green matrix $G_\Omega \in \mathbb{R}^{N\times N}$ is defined as the inverse (or pseudoinverse) of L_Ω,

$$G_\Omega = L_\Omega^{-1}, \quad u = G_\Omega f. \qquad (5)$$

Each column $G_\Omega(:, j)$ corresponds to the discrete Green function evaluated at grid points when a unit source is placed at x_j.

Standard neural operators such as FNO approximate the action of G_Ω on an input field f by composing spectral convolutions and pointwise nonlinearities:

$$u_\theta \approx \mathcal{T}_{\theta,\Omega}(f) = \mathcal{N}_L \circ \cdots \circ \mathcal{N}_1(f), \qquad (6)$$

where each layer $\mathcal{N}_\ell$ performs

$$\mathcal{N}_\ell(v)(x) = \sigma\Big(W_\ell v(x) + \mathcal{F}^{-1}\big(\hat{K}_\ell(k)\cdot \mathcal{F}[v](k)\big)(x)\Big), \quad (7)$$

with $\mathcal{F}$ the discrete Fourier transform and $\hat{K}_\ell$ a learned spectral kernel. When $\hat{K}_\ell$ does not depend on geometry, $\mathcal{T}_{\theta,\Omega}$ is geometry-agnostic and may implicitly encode an average operator across training domains.

In contrast, our GeoGreen-Op model (Section 3) explicitly parameterizes a geometry-conditioned approximation $G_\theta(x, y \mid \Omega)$ to $G_\Omega(x, y)$:

$$G_\theta(x, y \mid \Omega) \approx G_\theta^{\text{core}}(x-y \mid z_\Omega) + \sum_{r=1}^{R} a_r(x \mid h, z_\Omega)\, b_r(y \mid h, z_\Omega), \qquad (8)$$

where G_θ^{core} is defined via a geometry-conditioned spectral kernel, and the low-rank factors a_r, b_r capture non-stationary geometry effects.

In matrix form, if we collect $a_r(x_i)$ as diagonal matrices A_r and $b_r(x_j)$ as diagonal matrices B_r, the learned Green matrix approximates

$$G_\theta \approx G_\theta^{\text{core}} + \sum_{r=1}^{R} A_r B_r^\top, \qquad (9)$$

where G_θ^{core} is (approximately) Toeplitz and thus corresponds to a convolution operator. The spectral Green core is therefore closely related to the FNO kernel, while the low-rank correction adds non-stationarity and geometry adaptivity.

A.3 Linearity, Superposition, and Practical Deviations

Our formulation relies on an approximate linearity assumption: for a fixed geometry and carrier band, the mapping from sources to fields is close to linear, so that superposition holds:

$$u(x; \alpha_1 s_1 + \alpha_2 s_2) \approx \alpha_1 u(x; s_1) + \alpha_2 u(x; s_2). \quad (10)$$

In practice, non-linear hardware effects, saturation, and interference-management policies may introduce deviations. However, for typical planning and coverage scenarios at moderate load, the linear approximation is sufficiently accurate for REM purposes. Our experiments in Section 4 empirically validate that a linear Green-operator surrogate is expressive enough to match or outperform strong non-linear baselines while enabling better cross-geometry generalization.

B Model Architectures and Training Details

B.1 GeoGreen-Op Architecture Hyperparameters

We summarize the main architectural choices for GeoGreen-Op below. All hyperparameters were tuned on a held-out validation set.

Geometry encoder. We use a 2D U-Net-style encoder with four resolution levels. Each level consists of two 3×3 convolutions with GroupNorm and GeLU activations. The number of feature channels per level is $(32, 64, 128, 128)$. The local embedding $h(x)$ is taken from the second-highest resolution feature map and has dimension $C_h = 64$. The global latent $z_\Omega \in \mathbb{R}^{C_z}$ is obtained by global average pooling over $h(x)$ followed by an MLP with hidden size 128 and output dimension $C_z = 64$.

Source encoder. Transmitter height and band index are embedded with two separate learnable embeddings of dimension 8, concatenated and passed through a small MLP (hidden size 32, output 16). The resulting source embedding $e_s \in \mathbb{R}^{16}$ is broadcast to the grid and concatenated with geometry channels as additional input to the encoder.

Spectral core. We keep the lowest $K_x \times K_y$ Fourier modes with $K_x = K_y = 24$ for all models. The spectral kernel $\hat{K}_\theta(k \mid z_\Omega)$ is parameterized by an MLP with two hidden layers of size 64, taking as input the concatenation of sinusoidal-encoded frequency indices and z_Ω. We use $L = 4$ GeoGreen-Op blocks by default.

Low-rank correction. We set the rank to $R = 8$ in all main experiments. MLPs MLP_a and MLP_b both have two hidden layers of size 64, and output dimension R. For stability, we normalize a_r and b_r to have zero mean and unit variance over the grid before computing the correction in Eq. (17).

Readout and nonlinearity. Each block uses a 1×1 convolution $W^{(\ell)}$ to mix channel dimensions, followed by a GeLU nonlinearity. The final readout head consists of a 1×1 convolution mapping the last feature map to a single scalar channel (pathloss or RSSI) per grid cell.

B.2 Training Setup

Optimization. All models are trained with AdamW, using a base learning rate of 3×10^{-4}, weight decay 10^{-4}, and cosine learning-rate decay. We train for 200 epochs with batch size 8 per GPU on two NVIDIA A100 GPUs. We use gradient clipping with a max norm of 1.0.

Losses and normalization. We train on log-distance pathloss values, scaling each map to zero mean and unit variance per band. Unless otherwise stated, we set $\lambda_1 = 0.5$,$\lambda_2 = 0.5$ in the combined L_1–L_2 loss, and $\alpha = 0.5$ for the geometry-aware weighting term. LOS/NLOS masks and height bins for the weighting are derived from the URBAN-RM metadata.

Regularization. We apply dropout with rate 0.1 in the geometry encoder and within the spectral-core MLPs. For uncertainty experiments (Section 4.3), we additionally train ensembles of $M = 5$ GeoGreen-Op models with different random seeds.

B.3 Baselines and Implementation Details

We re-implemented or adapted the following baselines:

- **FNO** [Li *et al.*, 2020]: 2D FNO with four layers,32 channels, and the same number of Fourier modes as GeoGreen-Op. Geometry and source encodings are concatenated as input channels.
- **Geo-FNO** [Gao and Jin, 2021]: FNO variant with geometry-aware modulation. We condition the spectral kernels on a global geometry latent, similar to our spectral core but without low-rank correction.
- **DiffFNO** [Liu and Tang, 2025]: Diffusion-enhanced FNO with 4 denoising steps. We adapt the original code to 2D radio maps, using the same Fourier modes and channel sizes as FNO.
- **GNN-REM** [Chen *et al.*, 2023]: Graph-based REM with nodes corresponding to grid cells and edges defined by k-nearest neighbors in geometry space. We use a 3-layer graph neural network with 64 hidden units.
- **RadioFormer** [Fang and others, 2025]: Transformer encoder–decoder over flattened grids, with 8 heads and 4 layers. Positional encodings encode 2D coordinates and band indices.
- **Diffusion REM**: A conditional U-Net diffusion model similar to [Zhang *et al.*, 2024; Jia and others, 2025], with 4 downsampling blocks and 4 upsampling blocks. Conditioning is provided by geometry channels and source metadata.

All baselines are tuned to have a comparable number of parameters (10–20M) and trained with the same optimizer and schedule. Where official implementations are available, we used them as a starting point and verified that our reproduced performance matches reported numbers on public benchmarks within reasonable tolerance.

Table 1: RMSE (dB) for LOS vs. NLOS grid cells (all bands combined).

Method	LOS	NLOS
FNO	5.1	8.9
Geo-FNO	4.9	8.3
RadioFormer	4.7	8.0
Diffusion REM	4.8	8.1
GeoGreen-Op	**4.4**	**7.1**

Table 2: Ablation study on URBAN-RM (overall RMSE in dB).

Variant	RMSE
GeoGreen-Op (full)	**6.1**
w/o low-rank correction ($R = 0$)	6.5
w/o global latent z_Ω	6.6
12 Fourier modes (vs. 24)	6.7
48 Fourier modes (vs. 24)	6.2
geometry-agnostic FNO	7.0

C Additional Experimental Results

In this appendix we provide extended quantitative results that complement Section 4, including per-region and per-band metrics, geometry-stratified error statistics, ablation studies.

C.1 Geometry-Stratified Error Analysis

To better understand where the improvements come from, we stratify grid cells by LOS/NLOS status and by building height. Table 1 summarizes RMSE for LOS and NLOS pixels. GeoGreen-Op significantly reduces error in NLOS regions, where multiple reflections and diffractions dominate.

The performance breakdown further illustrates errors across height bins (low-rise, mid-rise, high-rise). We observe the largest relative gains in high-rise clusters, consistent with the design goal of geometry-aware Green kernels.

C.2 Ablation Studies

We perform ablations to isolate the contributions of geometry conditioning, the low-rank correction, and the number of Fourier modes. Table 3 shows RMSE averaged over all regions and bands.

The results indicate that both components of the Green kernel parameterization are important. Removing the low-rank correction or the global geometry latent degrades performance, and using too few spectral modes harms accuracy in complex geometries. Increasing the number of modes beyond 24 yields diminishing returns.

D Failure Cases and Qualitative Analysis

While GeoGreen-Op improves overall accuracy and generalization, it still exhibits systematic failure modes in certain edge cases. We summarize several representative patterns below. For each case, we have inspected representative tiles and error maps (plots omitted here for brevity).

D.1 Underestimated Diffraction Behind Sharp Corners

In some narrow street canyons with sharp 90° turns, the model tends to underestimate field strength immediately behind the corner, leading to overly pessimistic coverage predictions. In a typical example, ground-truth measurements show a weak but non-negligible diffraction tail extending a few tens of meters behind the corner, whereas GeoGreen-Op predicts a sharper drop with a "shadow" region that is too dark.

We hypothesize that this behavior stems from the limited spectral resolution and the approximate representation of high-frequency, non-stationary effects in the low-rank correction. Incorporating explicit diffraction-aware priors or higher-resolution local kernels could mitigate this issue.

D.2 Over-Smoothing in Highly Reflective Courtyards

In enclosed courtyards surrounded by mid-rise buildings, GeoGreen-Op occasionally over-smooths the standing-wave patterns observed in the measurements. Qualitatively, the predicted maps reproduce the overall attenuation trend but underestimate local peaks and nulls.

This suggests that the current kernel parameterization captures the dominant large-scale behavior but may be too smooth to reproduce fine-grained interference patterns. A possible remedy is to allocate more high-frequency spectral modes or introduce a small learnable correction focused on short-range geometry around the transmitter.

D.3 Sensitivity to Geometry Misalignment

When the building footprint or height map is slightly misaligned with respect to the true measurement grid (e.g., due to registration errors between GIS and measurement data), we observe localized artifacts near building edges. In such tiles, predicted coverage contours appear shifted by one or two grid cells relative to the ground truth.

This failure mode is not unique to GeoGreen-Op but is particularly visible because the learned Green kernel is strongly conditioned on geometry. Future work could incorporate data-driven alignment modules or explicit robustness augmentations that randomize small geometric shifts during training.

D.4 Attenuation Errors in Very High-Rise Clusters

In dense clusters of very high-rise buildings, the model sometimes underestimates penetration loss for receivers deep inside the cluster, leading to optimistic predictions of indoor or near-indoor coverage. Visual inspection shows that predicted fields decay with distance but do not fully capture the compounded attenuation from multiple layers of buildings.

This points to a limitation of our current 2.5D geometry encoding, which does not explicitly represent material properties or true 3D volumetric structures. Extending GeoGreen-Op to richer 3D geometry and material channels, or hybridizing with differentiable ray-tracing back-ends, is a promising direction to address this limitation.

E Dataset and Reproducibility Notes

E.1 URBAN-RM-Style Benchmark Organization

The URBAN-RM-style benchmark used in this paper follows recent practice in REM datasets [Yapar *et al.*, 2024; Zhang *et al.*, 2024]. At a high level, the dataset is organized as:

- `geometry/`: per-tile geometry tensors (building heights, land-cover, LOS/NLOS masks, etc.).
- `tx_meta/`: transmitter metadata (positions, heights, bands, powers).
- `maps/`: ground-truth pathloss or RSSI maps for each transmitter and band.
- `splits/`: train/validation/test split files listing tile and transmitter IDs for each split.

Each tile is stored as a separate file to facilitate sampling and parallel I/O during training. Geometry tensors share the same spatial resolution and alignment with the radio maps.

E.2 Train/Validation/Test Splits

We adopt three types of splits:

- **In-distribution**: randomly sample 70% of tiles for training, 10% for validation, and 20% for testing. Transmitters within a tile are partitioned so that no exact transmitter appears in both train and test.
- **Cross-geometry**: train on one subset of regions (e.g., city center + suburb) and test on a disjoint region (e.g., mixed residential). Validation tiles are drawn from training regions.

  All splits are defined via deterministic index files and are released together with the code to ensure reproducibility.

E.3 Reproducibility Checklist

We summarize key elements needed to reproduce our experiments:

- *Random seeds:* We fix seeds for Python, NumPy, and PyTorch. Experiments reported in the main text use seed=42 unless otherwise stated.
- *Hardware and software:* All models are trained on 2×A100 40GB GPUs with CUDA 12 and PyTorch 2.x. Training time per model ranges from 8 to 14 hours depending on architecture.
- *Hyperparameters:* All learning-rate schedules, batch sizes, number of epochs, and model dimensions are documented in configuration files that will be released with the code.
- *Evaluation:* We compute MAE, RMSE, and log-RMSE over all grid cells in the test split. Geometry-stratified metrics use mask files provided with the dataset.
- *Code and configs:* An anonymized repository containing training and evaluation scripts, model definitions, and configuration files will be made available upon publication (or as a supplementary artifact during review).

F Additional Qualitative Analysis under Sparse and Complex Geometries

This appendix provides additional qualitative visualizations that complement the quantitative results in Section 4. We focus on two challenging regimes that frequently arise in practical radio-environment-map (REM) construction: (i) sparse measurement settings near sharp geometric boundaries, and (ii) long-range propagation in dense and irregular urban layouts. These cases help elucidate how geometry-conditioned operators differ from geometry-agnostic neural operators beyond aggregate error metrics.

F.1 Boundary Behavior under Sparse Sampling

Figure 3 shows representative qualitative comparisons under sparse measurement settings, where only a limited number of observations are available near building edges and street boundaries. Each row corresponds to a different urban layout.

Across these examples, the FNO baseline exhibits noticeable granular artifacts and high-frequency noise, particularly near geometric discontinuities and in shadowed regions. Such behavior becomes more pronounced under sparse supervision, where spectral truncation and global convolutions tend to introduce spurious oscillations.

In contrast, the U-Net baseline produces overly smooth predictions. While large-scale attenuation trends are captured, sharp transitions across walls and narrow street canyons are blurred, leading to leakage across obstacles and loss of fine boundary structure.

GeoGreen-Op effectively combines the complementary strengths of these two baselines. Compared to FNO, it suppresses granular noise and spurious artifacts; compared to U-Net, it preserves sharp, boundary-aligned attenuation patterns. As a result, predicted fields remain well aligned with building contours, and transitions across walls are crisp and localized.

This behavior is consistent with the operator-theoretic perspective outlined in Appendix A: by explicitly parameterizing a geometry-conditioned Green operator, GeoGreen-Op is biased toward propagation patterns that respect boundary-induced anisotropy. Even with limited local observations, the learned operator extrapolates in a manner that is more consistent with physically plausible boundary interactions.

F.2 Error Distribution under Complex Urban Layouts

Figure 2 analyzes prediction errors aggregated over dense and irregular urban environments, with a particular emphasis on regions where long-range propagation and non-line-of-sight (NLOS) effects dominate. Rather than visualizing individual spatial maps, we report distribution-level statistics of per-pixel absolute errors to better characterize model behavior in challenging regimes.

The CDF curves show that GeoGreen-Op achieves a higher fraction of pixels with small absolute errors across a wide range of error thresholds. In particular, the inset reveals that Green-operator maintains a clear advantage in the medium-to-large error regime, suggesting a systematic reduction of

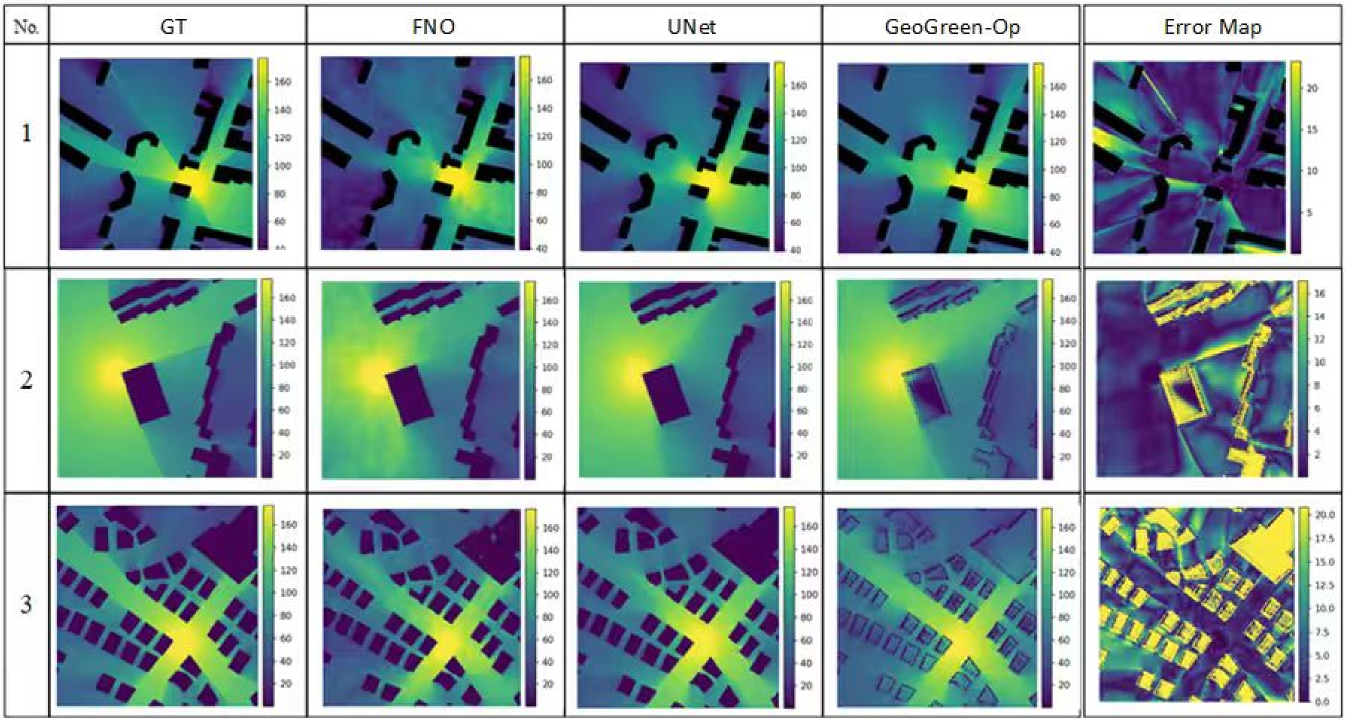

Figure 1: Boundary behavior under sparse sampling. From left to right: ground truth, FNO, U-Net, and GeoGreen-Op . FNO exhibits granular artifacts near geometric discontinuities, U-Net produces overly smooth transitions across obstacles, while GeoGreen-Op combines sharp boundary preservation with reduced spurious noise, yielding predictions that remain well aligned with building contours under sparse supervision.

heavy-tail errors that typically arise in complex propagation environments.

The error histograms provide a complementary view of the same phenomenon. Compared to FNO and U-Net, GeoGreen-Op concentrates substantially more probability mass near zero error, while exhibiting a noticeably lighter right tail. This indicates that the improvements of GeoGreen-Op are not driven by a small subset of easy pixels, but instead reflect a consistent reduction of large-error outliers across the domain.

Overall, these results demonstrate that geometry-conditioned Green operators improve not only average accuracy, but also the stability of predictions under long-range and structurally complex urban layouts, where error accumulation is a dominant challenge for geometry-agnostic models.

F.3 Discussion and Relation to Quantitative Results

These qualitative examples complement the quantitative gains reported in Section 4 by illustrating where geometry awareness matters most. In sparse and complex urban environments, accurate REM estimation requires not only fitting local statistics but also extrapolating across space in a physically plausible manner.

By explicitly conditioning the operator on urban geometry, GeoGreen-Op exhibits improved robustness near boundaries and reduced error accumulation in challenging long-range and non-line-of-sight regimes that are traditionally challenging for convolutional or geometry-agnostic neural operators. Together with the results in Appendix A, these visualizations support the view that learning geometry-conditioned Green operators provides a more faithful inductive bias for urban radio propagation.

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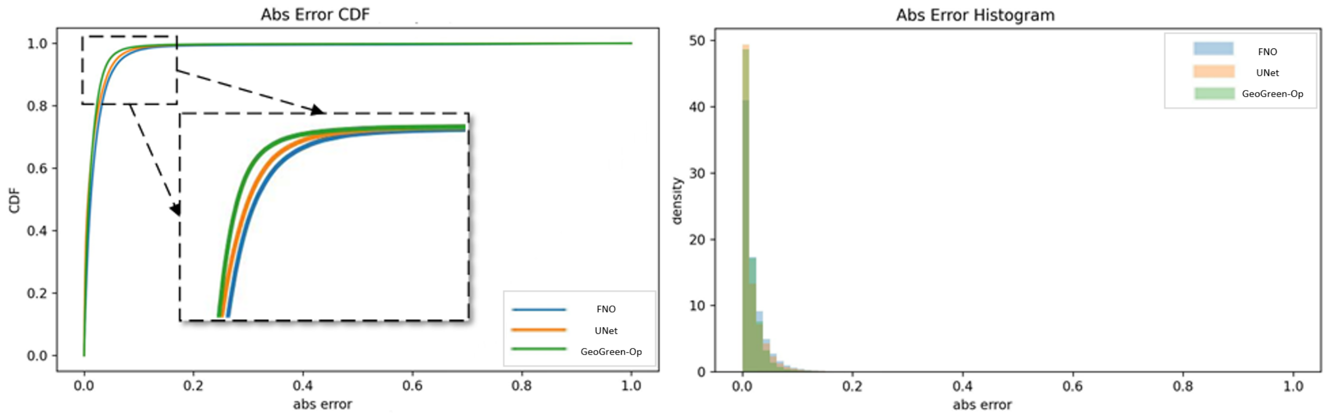

Figure 2: Distribution of per-pixel absolute prediction errors evaluated in the normalized prediction space and restricted to outdoor regions. *Left:* cumulative distribution function (CDF) of absolute errors, with an inset highlighting the medium-error regime. *Right:* corresponding error histograms shown as probability densities. GeoGreen-Op exhibits a consistently left-shifted CDF and a more concentrated error distribution near zero compared to FNO and U-Net, indicating fewer large-error outliers.

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